Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Reid, Michael B. Ph.D.		
Project Title:	Redox Modulation of Muscle Function in Microgravity		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Muscle Alterations and Atrophy Team		
Joint Agency Name:	National Institutes of Health/National Heart Lung and Blood Institute	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	michael.reid@uky.edu	**Fax:**	FY 859-323-1070
PI Organization Type:	UNIVERSITY	**Phone:**	859-323-6045
Organization Name:	University of Kentucky		
PI Address 1:	800 Rose Street, Room MS-509		
PI Address 2:			
PI Web Page:			
City:	Lexington	**State:**	KY
Zip Code:	40536-0298	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	04/01/2001	**End Date:**	07/31/2005
No. of Post Docs:	4	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Taylor, Addison (Baylor College of Medicine)		
Grant/Contract No.:	NCC 9-58-MA00209		
Performance Goal No.:			
Performance Goal Text:			

Exercise-induced fatigue and muscle atrophy are mediated in part by reactive oxygen species (ROS), a stimulus that may be exaggerated by radiation during spaceflight. The current project is assessing the roles of ROS signaling and radiation on muscle fatigue and atrophy and is testing antioxidants as possible countermeasures. Progress was hindered by severe damage to our institution by Tropical Storm Allison in June, 2001. These losses were resolved and we have made rapid progress on the project over the last year as outlined below: Aim 1. To determine if oxidative stress contributes to muscle fatigue during handgrip exercise. Fatigue of hand and forearm muscles may limit crew performance during extravehicular activity (EVA). N-acetylcysteine (NAC) is an antioxidant that inhibits muscle fatigue in humans. We have recently completed experiments testing the capacity of NAC to inhibit muscle fatigue and oxidative stress in humans during handgrip exercise. Working with Dr. Jeff Jones, Flight Surgeon at NASA Johnson Space Center, we used equipment and test procedures designed for use on the International Space Station. Results of the study show

Task Description:	the feasibility of this approach. NAC abolished glutathione oxidation blood draining the affected muscle groups and increased handgrip endurance by 30% relative to untreated trials. This aim has been completed and a manuscript is being prepared. Follow-up studies are being planned that will test the importance of these findings in a more operationally-relevant setting. Aim 2. To determine whether ionizing radiation accelerates ROS production and fatigue in skeletal muscle. We postulated that proton radiation absorbed during EVA would increase tissue ROS levels and accelerate muscle fatigue. We were testing this postulate in collaboration with Dr. Carlos Gonzalez, Director of the cyclotron at the University of Texas Medical School, when Tropical Storm Allison destroyed the cyclotron facility in 2001. The facility has not been rebuilt. Resources intended for these experiments have been redirected to studies of mechanisms regulating muscle atrophy and to tests of potential countermeasures (Aim 3). Aim 3. To evaluate oxidative stress as a mediator of muscle weakness caused by gravitational unloading. Muscle atrophy and contractile dysfunction cause weakness after prolonged spaceflight. We are evaluating oxidative stress as a cause of these changes in mouse soleus during 12-days of hindlimb unloading and are using cell culture techniques to evaluate mechanism. Our results show that: 1.) unloading increases oxidant activity within soleus muscle fibers; 2.) contractile dysfunction is blunted by administration of some antioxidants (NAC, allopurinol) but not others (curcumin, vitamin E); 3.) a novel ubiquitin conjugating enzyme, UbcH2/E220k, is highly expressed in skeletal muscle, is upregulated by ROS exposure, and mediates ubiquitin conjugation to muscle proteins; 4.) hydrogen peroxide upregulates expression of atrogin1/MAFbx, a key ubiquitin ligase that regulates muscle atrophy; 5.) this signal is transduced by p38 MAP kinase; and 6.) p38 inhibition blocks atrogin1/MAFbx upregulation and the associated rise in ubiquitin conjugating activity. We are testing the roles of these transcriptional mechanisms in atrophy of unloaded muscle and will continue to evaluate potential countermeasures. Aim 4. To determine if radiation stimulates atrophic signaling in muscle. We postulated that radiation-derived ROS might stimulate catabolic signaling and planned to measure activity of redox-sensitive pathways in muscle after proton irradiation. Destruction of the Medical Center cyclotron has prevented these experiments. Project resources have been redirected to studies of cellular mechanism and putative countermeasures (Aim 3).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research directly addresses two Earth-based problems, loss of function in unloaded muscle and muscle fatigue. The first problem occurs in individuals who are immobilized by injury or surgery. Muscles of the affected limbs atrophy and weaken, making it difficult for the individual to return to normal daily activity. The resulting inactivity lessens the quality of life, increases hospitalization and therapeutic costs, and increases the likelihood of pneumonia, venous thromboses, and other serious medical complications. A practical countermeasure to lessen atrophy and weakness would directly benefit these individuals, lessening the problems caused by transient immobilization. The second problem is familiar to us all. Acute muscle fatigue is a common feature of strenuous exercise. A countermeasure to inhibit fatigue would benefit a broad range of the US populace whose work requires physical exertion ranging from military professionals to firefighters, from police officers to construction workers. [The implications for professional athletes are all too obvious.]
Task Progress:	As stated previously, our data confirm that gravitational unloading causes oxidant activity to increase in mouse soleus fibers. The DCFH assay was used to test for increased oxidant activity in soleus muscle fibers of mice conditioned by hindlimb unloading for 12 days. Paired comparisons demonstrated increased oxidant activity in unloaded muscles (Matuszczak, et al. 2004). Pilot data suggest such increases can be inhibited by antioxidant supplementation. Mice conditioned by hindlimb suspension were fed diet containing Bowman-Birk Inhibitor Complex (BBIC) 1% or diet containing heat-inactivated BBIC. After 12 days, soleus muscles were isolated and oxidant activity was measured. Activity appears to be less in muscles of animals treated with BBIC. Antioxidant treatment also blunts the loss of specific force that occurs with unloading. Allopurinol, a xanthine oxidase inhibitor and nonspecific antioxidant protected soleus function in animals conditioned by 12 days' hindlimb unloading; across the physiological activation range of 0-80 Hz, specific forces developed by allopurinol-treated unloaded muscles were not different from control. Pilot data indicate specific force is also increased in unloaded soleus of mice that receive NAC dissolved in drinking water (+29%; $P<0.001$) or a diet containing 1% BBIC (+12%; $P<0.05$). Oxidative stress stimulates signaling events and transcriptional responses that favor muscle atrophy. These include activation of NF-kappaB and upregulation of components in the ubiquitin-proteasome pathway (Li, et al 2003). Interventions that diminish oxidant activity in unloaded muscle may depress catabolic signaling and inhibit atrophy. For example, curcumin (diferuloylmethane) inhibits NF-kappaB in cell culture systems and has nonspecific antioxidant properties. In pilot studies, we fed freely-ambulating mice a diet supplemented with 1% curcumin and found that it depressed NF-kappaB signaling in soleus, a response similar to the NF-kappaB inhibition produced by strenuous exercise (Durham, et al 2004). We postulate that prolonged NF-kappaB inhibition by curcumin or other antioxidants could inhibit atrophy of unloaded muscle. NAC is a nonspecific antioxidant that supports glutathione resynthesis and is reported to inhibit muscle fatigue. We tested NAC effects on performance of repetitive isometric handgrip exercise similar to the tasks performed by astronauts during extravehicular activity (EVA). Healthy volunteers were coached on task performance during practice sessions. In formal studies, we found that NAC administration inhibited glutathione oxidation during exercise and increased handgrip endurance by approximately 30% (Matuzszcak, et al., 2005).
Bibliography Type:	Description: (Last Updated: 08/24/2020)
Articles in Peer-reviewed Journals	Arbogast S, Reid MB. "Oxidant activity in skeletal muscle fibers is influenced by temperature, CO2 level, and muscle-derived nitric oxide. " Am J Physiol Regul Integr Comp Physiol. 2004 Oct;287(4):R698-705. Epub 2004 Jun 3. PMID: 15178539 , Oct-2004
Articles in Peer-reviewed Journals	Durham WJ, Li YP, Gerken E, Farid M, Arbogast S, Wolfe RR, Reid MB. "Fatiguing exercise reduces DNA-binding activity of NF-kappaB in skeletal muscle nuclei." J Appl Physiol. 2004 Nov;97(5):1740-5. Epub 2004 Jun 18. PMID: 15208298 , Nov-2004
Articles in Peer-reviewed Journals	Matuszczak Y, Farid M, Jones J, Lansdowne S, Smith MA, Taylor AA, Reid MB. "Effects of N-acetylcysteine on glutathione oxidation and fatigue during handgrip exercise." Muscle Nerve. 2005 Nov;32(5):633-8. PMID: 16025522 , Nov-2005
Articles in Peer-reviewed Journals	Matuszczak Y, Arbogast S, Reid MB. "Allopurinol mitigates muscle contractile dysfunction caused by hindlimb unloading in mice." Aviat Space Environ Med. 2004 Jul;75(7):581-8. PMID: 15267079 , Jul-2004

Articles in Peer-reviewed Journals	Reid MB. "Response of the ubiquitin-proteasome pathway to changes in muscle activity." Am J Physiol Regul Integr Comp Physiol. 2005 Jun;288(6):R1423-31. Review. PMID: 15886351 , Jun-2005
Articles in Peer-reviewed Journals	Farid M, Reid MB, Li YP, Gerken E, Durham WJ. "Effects of dietary curcumin or N-acetylcysteine on NF-kappaB activity and contractile performance in ambulatory and unloaded murine soleus." Nutr Metab (Lond). 2005 Aug 26;2:20. PMID: 16124875 , Aug-2005
Articles in Peer-reviewed Journals	Smith MA, Reid MB. "Redox modulation of contractile function in respiratory and limb skeletal muscle." Respir Physiol Neurobiol. 2006 Apr 28;151(2-3):229-41. Epub 2006 Feb 14. Review. PMID: 16481226 , Apr-2006
Books/Book Chapters	Moylan JS, Durham WJ, Reid MB. "Muscle, oxidative stress and aging." in "Oxidative stress, exercise, and aging." Ed. H.M. Alessio, A.E. Hagerman. London : Imperial College Press ; Hackensack, NJ : distributed by World Scientific Pub., c2006. ISBN: 9781860946196, 1860946194., Mar-2006
Presentation	Reid, M. B. "Can Antioxidants Improve Muscle Performance? " Dec-2002
Presentation	Reid, M. B. "Exercise Regulation of the Ubiquitin-Proteasome System " May-2004
Presentation	Reid, M. B. "Redox Modulation of Muscle Function in Microgravity " Jan-2004
Presentation	Reid, M. B. "Regulation of Muscle Adaptation by Redox Signaling " May-2004